

Neural Network Assignment

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1. Role of Weights in a Neuron

Weights are numerical values assigned to each input in a neuron. They determine the importance or influence of each input feature on the output. During training, these weights are adjusted to minimize error and improve the model's predictions. A higher weight means the input has more impact on the output, while a lower weight means less impact. Weights are learned automatically using optimization algorithms like gradient descent.

2. What is an Activation Function?

An activation function is a mathematical function applied to the output of a neuron to introduce non-linearity into the model. It helps the neural network learn complex patterns and relationships in the data. Without activation functions, the network would behave like a simple linear model. Common activation functions include Sigmoid, ReLU (Rectified Linear Unit), and Tanh.

3. Define Probability Distribution in ML Context

A probability distribution describes how the values of a random variable are distributed and how likely each value is to occur. In machine learning, probability distributions are used to model uncertainty, make predictions, and estimate outcomes. For example, classification models often output probabilities indicating the likelihood of each class. These distributions help in decision-making and evaluating model confidence.

4. What is Gradient in Optimization?

The gradient is a vector of partial derivatives that represents the direction and rate of change of a function. In machine learning optimization, the gradient indicates how the model parameters (like weights) should be adjusted to minimize the loss function. Algorithms like gradient descent use the gradient to update weights in the direction that reduces error, helping the model learn efficiently.
